The oracle risk of linear shrinkage

Abstract

An observation X with mean θ and variance σ^2 is shrunk towards zero by a factor c. We compute the best factor and its risk under squared error, and compare that risk with the risks of the two simplest estimators, 0 and X.

1 Setting

Throughout, $\theta \neq 0$ and $\sigma > 0$ are real numbers.

Definition 1.1. For $c \in \mathbb{R}$, the *risk* of the estimator cX is $R(c) = c^2\sigma^2 + (1-c)^2\theta^2$, its mean squared error when X has mean θ and variance σ^2. The *oracle factor* is $c^* = \theta^2/(\theta^2+\sigma^2)$. The *oracle risk* is $\rho = \theta^2\sigma^2/(\theta^2+\sigma^2)$. It is the risk of the oracle factor, as Lemma 2.1 shows.

2 Results

Lemma 2.1. *For every $c \in \mathbb{R}$, $R(c) - \rho = (\theta^2+\sigma^2)(c-c^*)^2$.*

Proof. Write $s = \theta^2+\sigma^2$, so that $c^* = \theta^2/s$ and $1-c^* = \sigma^2/s$. For $c = c^* + d$,

$$R(c) = (c^*+d)^2\sigma^2 + (1-c^*-d)^2\theta^2 = \frac{\theta^4\sigma^2+\sigma^4\theta^2}{s^2} + 2d\Big(\frac{\theta^2\sigma^2}{s} - \frac{\sigma^2\theta^2}{s}\Big) + s\,d^2 = \rho + s\,d^2,$$

which is the claim. □

Lemma 2.2. $\rho \leq \min\{\theta^2, \sigma^2\}$.

Proof. Since $\theta^2+\sigma^2 > \sigma^2 > 0$, we have $\rho = \theta^2\cdot\sigma^2/(\theta^2+\sigma^2) < \theta^2$, and in the same way $\rho < \sigma^2$. □

Theorem 2.3. *For every $c \in \mathbb{R}$, $R(c) \geq \rho$, with equality if and only if $c = c^*$. Moreover, $\rho \leq \min\{R(0), R(1)\}$.*

Proof. By Lemma 2.1, $R(c) - \rho = (\theta^2+\sigma^2)(c-c^*)^2 \geq 0$, with equality if and only if $c = c^*$, because $\theta^2+\sigma^2 > 0$. Since $R(0) = \theta^2$ and $R(1) = \sigma^2$, Lemma 2.2 gives $\rho \leq \min\{R(0), R(1)\}$. □

Corollary 2.4. $R(c^*) \leq R(1/2)$.

Proof. By Lemma 2.1, $R(1/2) - R(c^*) = (\theta^2+\sigma^2)(1/2-c^*)^2 \geq 0$. By Lemma 2.2, moreover, $R(c^*) = \rho \leq \min\{\theta^2, \sigma^2\}$. □